

Yonghoon Jeong

Republic of Korea Naval Academy

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Research Interests

Wireless Communication Systems
Performance Analysis and Modeling
Physical-Layer Wireless Sensing and Networking
Low-Power Wide-Area Networks (LPWANs)
Ocean Internet of Things (Ocean IoT)

Education

M.S. in Electrical and Computer Engineering Seoul National University, Seoul, Republic of Korea Advisor: Prof. Saewoong Bahk (currently President of ETRI)	2024 - 2026
B.S. in Electronics and Communications Engineering Kwangwoon University, Seoul, Republic of Korea	2018 - 2023

Professional Experience

Republic of Korea Naval Academy Faculty Member, Department of Cyber Science Researcher, Institute of Naval Research Commissioned as a Republic of Korea Navy officer. Plan to pursue a Ph.D. abroad following military service completion in May 2029.	2026 - Present
Seoul National University (SNU) Researcher, Network Laboratory (NETLAB), Institute of New Media and Communications (INMC)	2024 - 2026
Electronics and Telecommunications Research Institute (ETRI) Researcher, AI-SoC Infrastructure Research Section	2023
Korea Advanced Institute of Science and Technology (KAIST) Research Intern, Unmanned Systems Research Group, School of Electrical Engineering Advisor: Prof. David Hyunchul Shim	2022

Publications

International Papers

- "On the Symbol Error Rate of Underwater Acoustic Backscatter Systems." **Under review** at *IEEE Transactions on Communications*.
- "Underwater Backscatter for Ocean Sensing: From Wireless Power to Data Collection." **Under review** at *IEEE Wireless Communications* (Magazine).
- "Characterizing the Relation Between the Energy Scattering Effect and Symbol Error Rate in Non-Linear Chirp Modulation for LoRa." **Under review** at *IEEE Wireless Communications Letters*.
- "Symbol Error Rate Analysis and Approximation for Arbitrary Chirp Modulation Under Multipath, Doppler, and Interference." **Under review** at *IEEE Transactions on Communications*.
- "On the Symbol Error Rate of Non-Linear Chirp Modulation for LoRa Under Asynchronous Interference." **Under review** at *IEEE Wireless Communications Letters*.
- **Yonghoon Jeong**, Youngwook Son, and Saewoong Bahk, "On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect." **Published** in *IEEE Communications Letters*, vol. 30, pp. 2238-2242, Jun. 2026. ([Paper](#))
- Dongrack Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, "D²-PLC: Holistic Design and Implementation of High-Data-Rate Duplex DC Power Line Communication Network." **Published** in *IEEE Internet of Things Journal*, vol. 12, pp. 45325-45340, Nov. 2025. ([Paper](#))

- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, “Demo: A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration.” **Published** in the proceedings of *IEEE ICNP 2025*, Seoul, Republic of Korea, Sep. 22-25, 2025. **(Paper)**
- **Yonghoon Jeong** and Youngseok Choi, “Diffusion-Phys: Noise-Robust Heart Rate Estimation from Facial Videos via Diffusion Models.” **Published** in *Biomedical Engineering Letters*, vol. 15, pp. 575-585, Apr. 2025. **(Paper)**

Domestic Papers

- **Yonghoon Jeong** and Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation.” **Published** in the proceedings of *The Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Republic of Korea, Nov. 20-22, 2024.
- **Yonghoon Jeong**, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video.” **Published** in the proceedings of *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Republic of Korea, Jun. 23-25, 2021.

Academic Service

Journal Reviewer

IEEE Internet of Things Journal

Awards & Scholarships

Scholarship Recipient	2024 - 2025
Brain Korea 21 (BK21), National Research Foundation of Korea	
Scholarship Recipient	2022 - 2023
Korea Scholarship Foundation for the Future Leaders (KOSFFL)	
Excellent Paper Award	2021
Korea Computer Conference 2021	
University President's Award	2020
Seoul Capstone Design, Seoul	
Scholarship Recipient	2020
Samsung Scholarship Foundation	
Dean's List	2020
Highest GPA	

Teaching Experience

Creative Engineering Design	2024
Teaching Assistant; Prof. Saewoong Bahk	
Electronic Circuits Experiment I	2023
Instructor; Emeritus Prof. Young-chul Chung	
Electronic Circuits Experiment II	2023
Instructor; Emeritus Prof. Young-chul Chung	
Electronic Circuits Experiment I	2022
Instructor; Prof. Youngseok Choi	

Technical Skills

Programming Languages: C, C++, Python, MATLAB

Frameworks & Libraries: ROS, ROS2, TensorFlow, PyTorch

Simulation Tools: MATLAB/Simulink, ns-3

Development Tools: Linux, Git, Docker